

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 6 Microbial Regulatory Systems

6.1 Multiple Choice Questions

- 1) Regulation of enzyme activity occurs
A) at the start of transcription.
B) at the start of translation.
C) posttranslationally.
D) at any point on the enzymatic production pathway.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 7.15

- 2) When arginine is added to a culture already growing exponentially in a medium without arginine, what occurs?
A) All cellular growth ceases.
B) Growth continues, but the production of enzymes required for the synthesis of arginine ceases.
C) Growth continues, but the production of enzymes required for the synthesis of arginine increases.
D) The cell returns to the lag stage of growth to synthesize the proteins necessary for the metabolism of arginine.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.2

- 3) Regulatory proteins
A) are influenced by small molecules.
B) bind to specific DNA sites.
C) regulate transcription.
D) regulate transcription, bind specific DNA sites, and can be influenced by small molecules.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

- 4) A protein region with a specific function and structure is called a
A) conserved site.
B) domain.
C) locale.
D) motif.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

5) Transcriptional regulators bind most frequently at the _____ site of DNA.

- A) major groove
- B) minor groove
- C) histone complex
- D) primary supercoil

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

6) Which type of regulator(s) specifically binds to operator regions of DNA?

- A) activators
- B) activators and inducers
- C) repressors
- D) repressors and corepressors

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

7) The *lac* operon is an example of _____ control in which the presence of an _____ is required for transcription to occur.

- A) negative / activator
- B) negative / inducer
- C) positive / activator
- D) positive / inducer

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

8) Enzyme induction occurs

- A) when the substrate is present.
- B) when the organism is environmentally stressed.
- C) continuously.
- D) when the substrate is depleted.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

9) Considering the catabolite repression mechanism, which observation would make you suspect it **IS** occurring?

- A) CRP bound to promoter sites
- B) elevated levels of transcripts for maltose and sucrose catabolism
- C) relatively low intracellular cyclic AMP levels
- D) RNA polymerase bound to biosynthetic promoter sequences

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.4

10) During a growth curve of *Aliivibrio fischeri*, when would you expect to see the strongest bioluminescence?

- A) lag phase
- B) early to middle log phase
- C) late log to early stationary phase
- D) middle to late stationary phase

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.8

11) What occurs when an inducer is added to an environment containing an organism with a metabolic pathway controlled by a repressor?

- A) The inducer combines with the repressor and activates the pathway.
- B) The inducer combines with the repressor and inactivates the pathway.
- C) The inducer combines with the substrate and blocks induction.
- D) The inducer does not combine with, but functions as a chaperone molecule for, the enzyme-substrate complex.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.2

12) Which of the following do NOT bind to the promoter sequence during regulation?

- A) activators
- B) inducers
- C) repressors
- D) None of these bind directly to the promoter sequence.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

13) Cyclic AMP is synthesized from ATP by an enzyme called _____ which is involved in _____.

- A) adenylate cyclase / catabolite repression
- B) adenylate cyclase / transcriptional activation
- C) cAMP receptor protein (CRP) synthase / catabolite repression
- D) cAMP receptor protein (CRP) synthase / transcriptional activation

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.4

14) In negative control of transcription by the *lac* operon, how does the presence of an inducer affect transcription?

- A) The inducer binds to the operator.
- B) The inducer does not bind to the operator.
- C) The inducer causes the repressor to bind to the operator.
- D) The inducer prevents the repressor from binding to the operator.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

15) Mechanisms for controlling enzyme activity include all of the following EXCEPT

- A) degradation of the enzyme.
- B) feedback inhibition.
- C) covalent modification of the enzyme.
- D) addition of short sections of new amino acid sequence.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

16) The function of a kinase is

- A) methylation.
- B) response regulation.
- C) phosphorylation.
- D) glycosylation.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.6

17) The promoters of positively controlled operons require activator proteins because

- A) RNA polymerase easily recognizes the consensus sequence.
- B) they are required to inactivate the repressor proteins.
- C) the promoters bind RNA polymerase weakly and utilize activator proteins to help RNA polymerase recognize the promoter.
- D) they are needed to bind to the allosteric site of RNA polymerase.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.3

18) Transcriptional control in Archaea most closely resembles that in _____ and utilizes _____.

- A) bacteria; regulatory proteins
- B) eukaryotes; regulatory proteins
- C) bacteria; transcription factors
- D) eukaryotes; transcription factors

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.5

- 19) The most frequent way in which small regulatory RNA molecules exert their effects is by
- A) base pairing with other RNA molecules that have regions of complementary sequence.
 - B) binding to a repressor and repressing enzyme transcription.
 - C) acting as an inducer that then binds to an activator protein to allow transcription to proceed.
 - D) signal transduction.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.11

- 20) _____ pathways typically rely on _____ proteins to inhibit mRNA synthesis.

- A) Anabolic; repressor
- B) Catabolic; repressor
- C) Anabolic; activator
- D) Catabolic; activator

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

- 21) _____ pathways typically utilize _____ proteins that stimulate binding of RNA polymerase to DNA.

- A) Anabolic; repressor
- B) Catabolic; repressor
- C) Anabolic; activator
- D) Catabolic; activator

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

- 22) In *Bacteria*, sensor kinases that respond to extracellular signals transfer this signal to the cytoplasmic machinery by typically phosphorylating the residues.

- A) histidine
- B) serine
- C) threonine
- D) tyrosine

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.6

23) Quorum sensing is a regulatory system that requires a certain cell density to work effectively. Each of the following activities utilizes quorum sensing EXCEPT

- A) *Staph aureus* infection.
- B) transition of *Candida albicans* from budding yeast to elongated filaments.
- C) flagella synthesis in *Proteus vulgaris*.
- D) light emission by *Aliivibrio fischeri*.

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.8

24) Chemotaxis in bacteria occurs through the use of

- A) adaptation.
- B) quorum sensing.
- C) autoinduction.
- D) a modified two-component system.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.7

25) Which statement is TRUE of two separate regulators controlling one individual operon?

- A) The two regulators themselves must respond to different signals, which enables both to control the operon differently.
- B) One regulator will likely control the transcription of one section of the operon, whereas the other regulator will control the other component.
- C) One regulator will bind to the operator region whereas the other will bind to the promoter region so they can co-occur and co-regulate the operon.
- D) Two regulators trying to control the same operon will likely result in only one being maintained after several generations.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.4

26) How would you design an sRNA to bind to a sequence?

- A) select six continuous nucleotides from the sequence
- B) take the complementary sequence of six continuous nucleotides
- C) select 200 continuous nucleotides from the sequence
- D) take the complementary sequence of 200 continuous nucleotides

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.11

- 27) When the nontemplate strand of a gene is transcribed into RNA, what is likely to result?
- A) A complementary sRNA will bind to it and form a functional ribozyme with secondary structure.
 - B) It will complementarily bind to the gene sequence, form a hairpin loop, and transcriptionally repress the gene.
 - C) The complementary mRNA transcribed from the template strand will bind to it and halt its translation.
 - D) A global regulator will identify this as a stress, respond by inducing ribonuclease production, and it will be degraded.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.11

- 28) Based on our understanding of the early stages of life, _____ is/are thought to be one of the earliest forms of metabolic regulation that evolved.

- A) attenuation
- B) feedback inhibition
- C) riboswitches
- D) transcription factors

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.12

- 29) Attenuation is a type of regulation that can control

- A) allosteric enzyme activity.
- B) transcriptional activity exclusively.
- C) translational activity exclusively.
- D) both transcriptional and translational activity.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.13

- 30) Which of the following is a characteristic of an isoenzyme?

- A) More than one enzyme is regulated by the same mechanism.
- B) The same reaction can be catalyzed by multiple enzyme variants.
- C) Multiple binding sites on the same enzyme enable multiple regulation mechanisms.
- D) More than one gene makes the same enzyme.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.14

31) Which regulatory mechanism does NOT depend on a conformational change in protein/enzyme structure to change activity?

- A) attenuation
- B) catabolite repression
- C) feedback inhibition
- D) negative control

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.13

32) An organism grown in nutrient rich broth to high turbidity always produces a blue pigment. When a large inoculum is transferred to a nutrient rich agar plate, it also appears blue. A researcher noticed that this never happens when very small colonies are grown on low nutrient agar plates, however. What is the most plausible conclusion?

- A) Large populations enabled the differentiation of a subpopulation of cells that created the blue pigment.
- B) Only high nutrient conditions provide enough energy for cells to produce this secondary metabolite that appears blue.
- C) Production of the blue pigment production is linked to quorum sensing.
- D) The strong gradient from very high to low nutrient bioavailability induces production of the blue metabolite.

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.8

33) A bacterium that either partially or fully catabolizes an acyl-homoserine lactone will likely disrupt

- A) attenuation.
- B) chemotaxis.
- C) endospore formation.
- D) quorum sensing.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.8

34) Phosphorylation of _____ regulates which direction a flagellum rotates, thus controlling whether an organism runs or tumbles.

- A) CheAW
- B) CheB
- C) CheY
- D) CheZ

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.7

35) Which organism would likely harbor the most two-component regulatory systems?

- A) an archaeon living in an extreme environment
- B) a bacterium occupying a heterogeneous niche with high nutrient mixing
- C) an organism capable of quorum sensing
- D) a parasitic bacterium living inside another organism

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.6

36) How is the activity of a riboswitch controlled?

- A) by other riboswitches
- B) metabolite binding can change its structure
- C) sigma factor binding alters its structure
- D) small RNA complementary binding disrupts its function

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.12

37) A mutation in the gene encoding the lactose repressor (lacI) that prevents lactose from binding to the LacI protein would result in

- A) constant expression of the lac operon in the absence of lactose.
- B) constant repression of the lac operon in the absence of lactose.
- C) constant expression of the lac operon in the presence of lactose.
- D) constant repression of the lac operon in the presence of lactose.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.4

38) Why are two-component regulatory systems particularly useful for controlling gene expression in response to environmental signals?

- A) Two proteins controlling a gene means there are two chances to activate a gene.
- B) Two proteins delay the response time so the cell can be sure the change is permanent.
- C) One of the two proteins can be exposed to the external environment to receive a signal.
- D) Phosphorylation is a permanent change so genes are always turned on after signal.

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.6

39) In order for the helix-turn-helix motif to bind to DNA, the _____ must fit into the major groove of the DNA.

- A) homeotic switches
- B) recognition helix
- C) operator
- D) zinc fingers

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

40) Infection by the foodborne pathogen *E coli* O157:H7 involves all of the following host and pathogen signaling molecules EXCEPT

- A) AI-3.
- B) MCP.
- C) epinephrine.
- D) norepinephrine.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.8

41) Global regulatory systems include all of the following EXCEPT

- A) amino acid synthesis.
- B) transformation.
- C) heat shock response.
- D) anaerobic respiration.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.10

42) When more than one operon is under the control of a single regulatory protein, the operons are collectively called a(n)

- A) regulon.
- B) operator.
- C) autoinducer.
- D) riboswitch.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.3

43) Most riboswitches control

- A) transcription.
- B) translation.
- C) ribosome synthesis.
- D) tRNA synthesis.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.12

44) PII proteins are a family of signal-transducing proteins that regulate _____ during nitrogen metabolism.

- A) transcription factors.
- B) enzymes.
- C) membrane transport proteins.
- D) PII proteins regulate each of the above during nitrogen metabolism.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.15

45) All of the following are functions of heat shock proteins in bacteria EXCEPT

- A) prevention of inappropriate protein subunit aggregation.
- B) degradation of denatured proteins.
- C) responding to exposure to high levels of ethanol.
- D) stimulation of binary fission.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.10

6.2 True/False Questions

1) Proteins required at approximately the same level throughout a cell's growth cycle are often not subject to regulatory mechanisms and are constitutively synthesized.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

2) Short regions at the beginning and end of gene sequences are not translated into proteins.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

3) Small molecules usually act directly (rather than indirectly) in regulating transcription.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

4) A common structure for proteins that bind DNA is helix-turn-helix.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

5) The residues in a DNA-binding protein that interact with DNA usually correspond to each other according to the amino acids encoded by the DNA. For example, a DNA sequence containing AGC-AGA-CAG which encodes for Ser-Arg-Gln would likely have a DNA-binding protein with the sequence Ser-Arg-Gln.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.1

6) Some proteins that bind to DNA block transcription, whereas other proteins activate transcription.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

7) Enzyme repression typically affects catabolic pathways.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.2

8) Depending on the type of regulatory mechanism, activators and repressors can bind to operator regions which control transcription.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.2

9) A regulon is composed of at least two operons regulated by the same protein.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.3

10) The preferential use of glucose over other available carbon substrates for growth is mechanistically explained by catabolite repression.

Answer: TRUE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.4

11) Proteins and RNA molecules that are needed in the cell at about the same level under all growth conditions, require constitutive expression.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.1

12) A two-component regulatory system usually involves both the sensor and response proteins being subject to phosphorylation.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.6

13) Cyclic AMP (cAMP) is involved in the global regulation of catabolic pathways in *Escherichia coli*, including the *lac* operon. A high activity of the enzyme adenylate cyclase which makes cAMP is suggestive of catabolite repression.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.4

14) Once cytoplasmic sensor proteins involved in chemotaxis regulation are phosphorylated, a cascade of other phosphate-transferring mechanisms provides ATP to rotate flagella.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.7

15) In most cases, the first product of a particular biosynthetic pathway represses the enzymes of the pathway.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.14

16) In catabolic repression, cells always use the least abundant carbon source first.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.4

17) The small size of prokaryotic cells limits their ability to respond to spatial gradients of a chemical.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.7

18) Adaptation is based on the level of phosphorylation of MCPs.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.7

19) Quorum sensing relies upon a large cell population which then turns on transcription.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.8

20) During chemotaxis, attractants increase the rate of autophosphorylation, whereas repellants decrease this rate.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.7

21) Heat shock proteins are produced in high number to minimize damaging effects during high heat conditions, in addition to other stress signals unrelated to heat such as ultraviolet radiation.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.10

22) Regulation of transcription in *Archaea* is more similar to that which occurs in eukaryotes than in bacteria.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.5

23) The stringent response is a mechanism used by bacteria that provides increased ability to survive in environments lacking nutrients.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.9

24) When environmental levels of amino acids become limited, synthesis of mRNA, rRNA, and tRNA increases immediately.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.9

25) In many environments, the limiting nutrient for bacterial growth is phosphorus.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.10

26) Multiple mechanisms for small RNA (sRNA)-mediated translational regulation exist, but all are unified by inactivating protein synthesis of the target mRNA.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.11

6.3 Essay Questions

1) Explain a common mechanism used by cells to ensure proper binding of DNA-binding proteins to DNA.

Answer: DNA-binding proteins are proteins composed of DNA-binding domains and that have a specific or general affinity for either single or double stranded DNA. One of the most common domains is the *helix-turn-helix structure* which consists of two segments of polypeptide chain that have an α -helix secondary structure connected by a short sequence forming the "turn." The first helix is the recognition helix which interacts specifically with the DNA. The second helix is the stabilizing helix which stabilizes the first helix by interacting with it through hydrophobic interactions. The turn linking the two helices consists of three amino acids with glycine typically the first. Sequences are recognized by noncovalent interactions between the recognition helix of the protein and specific chemical in the sequence of base pairs on the DNA.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.1

2) Using the tryptophan operon as a model, describe the formation of a stem-loop structure and its function.

Answer: A pause in mRNA transcription occurs when tryptophan is in short supply and therefore requires the additional recruitment of a charged tryptophan tRNA for its synthesis. This lag or pause time provides the already synthesized mRNA component to bind to a different part of the transcript than would occur if there were no pause time. This different hairpin loop formed does not interact with the DNA sequence and therefore permits RNA polymerase to continue transcribing the entire mRNA, unlike what would occur if a lag time were absent.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.13

3) Explain the phenomenon known as diauxic growth. Hypothesize why this type of regulation might be important for the survival of a microbe in a mixed population.

Answer: Consider *Escherichia coli* as an example: when *Escherichia coli* is exposed simultaneously to both glucose and lactose, it will preferentially use glucose before lactose. Diauxic growth therefore explains how cells control gene expression, often as operons and regulons, to achieve the fastest growth rate. If a cell were to not have this mechanism, it might be outcompeted in the environment by other faster growing and replicating cells. Lacking a global catabolite regulation system might be advantageous for cells occurring in a mixed population where competition for a single substrate could provide poor growth yields.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.4

4) Describe the basic two-component regulatory system, and explain the function of each component.

Answer: The first component of the system involves a specific sensor kinase protein located in the cytoplasmic membrane which receives an external environmental signal. The kinase autophosphorylates itself once signaled and creates a phosphate ion for the second component. A response regulator protein within the cell's cytoplasm then binds the phosphate ion and ultimately binds DNA. The regulation in response to this DNA binding can either be positive or negative, depending on the system.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.6

5) Explain why is it unsurprising that two-component regulatory systems are all but absent in bacteria that live at the expense of a host.

Answer: Answers will vary, but a possible explanation is that an organism that has adapted to living inside of a host has such a narrow metabolic capacity that responses to nutrient limitation are less necessary than a bacterium growing in a dynamic community.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.6

6) Explain why one cell of a pathogenic bacterium by itself typically does not secrete a toxin, despite the species as a whole being characterized as toxin producing. To help in your explanation, predict what would happen to an individual pathogen that secreted a toxin inside a human.

Answer: One pathogenic bacterium is unlikely to be able to produce a sufficient number of toxin molecules to cause a disease. Once the bacterium grows to a sufficient quantity that it can cause disease through the production of toxins, coordination of toxin production by the population is performed by quorum sensing. The production of a toxin by a single cell would likely alert the immune system a foreign bacterium is present and would be killed by a healthy immune system.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.8

7) How is transcription in *Archaea* controlled?

Answer: *Archaea* have mechanisms to control transcription, which are most similar to those found in *Bacteria* rather than other mechanisms such as those in *Eukarya*. They contain both activators and repressors, which can regulate RNA polymerase activity in either a positive or negative way.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.5

8) Contrast prokaryotic and eukaryotic regulation of gene expression.

Answer: In *Bacteria* and *Archaea*, a common mechanism for regulation of RNA polymerase activity is to use DNA-binding proteins that either block RNA polymerase activity (repressor proteins) or stimulate RNA polymerase (activator proteins). An alternative mechanism found commonly in eukaryotes is to coordinate numerous DNA-binding proteins known as transcription factors to interact with RNA polymerase.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.5

9) Explain the function of cAMP in catabolite repression.

Answer: Adenylate cyclase synthesizes the regulatory nucleotide cAMP, and when the substrate glucose is present this process is inhibited (or repressed). This means glucose is a preferred substrate for catabolism in cells that contain this mechanism, such as *E. coli*. Once glucose has been used up, cAMP is then produced, which forms a cAMP—CAP—DNA complex, resulting in transcription of the genes that the CAP corresponds to, and catabolism of an additional substrate can be performed.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.4

10) A cocktail of 30 compounds was identified to harbor a corepressor, and subsequent separation and purification was performed to test individual compounds. Describe an experiment that would enable you to identify the specific compound that acts as a corepressor for a catabolic pathway. Be certain to explain what will be measured and how you could conclude which compound is a corepressor.

Answer: Experimental designs will vary, depending on whether transcripts, proteins, or enzyme function is quantified. Measurement of each would enable measuring activity of the repressor. In each case, a compound added to the system which shows significantly lower (basal) activity would indicate it is a corepressor. Compounds that do not act as a corepressor would result in no change in activity, meaning transcripts, proteins, and enzyme function would still be observed.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.2

11) In considering the function of heat shock proteins, why is it not surprising that these proteins are both highly conserved and very ancient? Provide your reasoning.

Answer: Answers will vary, but one idea is for an organism to survive in the environment, it must be able to survive unfavorable conditions. Recognizing that heat shock proteins generally facilitate the refolding or removal of misfolded proteins, whether it be from heat or other conditions, it makes sense these are so critical for a cell to survive that they would have needed to be present even in ancient organisms. Additionally, whether a protein is misfolded in a eukaryote or a prokaryote, it seems likely that a heat shock protein would be needed to refold it. These proteins may be similar, which explain the high conservation throughout all life forms.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.10

12) Provide evidence that supports the hypothesis that riboswitches are remnants of the RNA world when catalytic RNAs were the only self-replicating life forms.

Answer: Answers will vary, but one focus could be on their presence in bacteria that are considered to have evolved well before *Eukarya*, yet riboswitches are generally lacking in eukaryotes. If no other life forms were around and RNA was the only replicating form, the function of riboswitches could explain a (primitive) mechanism to control its own synthesis.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.12

13) Compare and contrast chemical, spatial, and temporal gradients as found in bacterial regulatory systems.

Answer: Bacteria cannot sense spatial gradients of chemicals due to their small size, so they are capable only of determining a chemical gradient over time. That is, they can sense the change in concentration of a chemical over time rather than the absolute concentration of the chemical stimulus. These temporal changes of a chemical are sensed with a two-component system. In the example of chemotaxis, an attractant or repellent's presence results in a change in activity of pre-existing flagella.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 6.7

14) Two common levels of regulation of protein activity are transcriptional control and posttranslational control. Discuss the differences between the two in terms of speed of response and energy efficiency.

Answer: Transcriptional control will be the slowest (the gene needs to be transcribed, the mRNA translated, and the protein correctly folded before protein activity) but is the most energy efficient because proteins are only made when they are needed. Posttranslational control is the fastest (the protein is already synthesized; it just needs a modification) but is less energy efficient because a protein might be made and degraded before it ever gets activated.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.14

15) Many of the virulence factors produced by the opportunistic pathogen *Pseudomonas aeruginosa* are the result of quorum sensing. What role do autoinducers play in this mechanism of bacterial pathogenicity?

Answer: Through the use of autoinducers, bacteria can regulate their behavior according to population density. The phenomenon of quorum sensing relies on the principle that when a single bacterium releases autoinducers (AIs) into the environment, their concentration is too low to be detected. However, when sufficient bacteria are present, autoinducer concentrations reach a threshold level that allows the bacteria to sense a critical cell mass and, in response, to activate or repress target genes including those of various virulence factors.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.8

16) Describe the difference between constitutive and inducible genes. Based on your knowledge of bacterial cells, give an example of each.

Answer: Constitutive genes are those which are transcribed at a relatively constant level regardless of the cell's environmental conditions. These genes are sometimes referred to as 'housekeeping genes' and include genes for heat shock proteins such as *dnaK* in *Escherichia coli*. The expression of inducible genes is highly regulated and these genes must be able to be rapidly and specifically activated in response to environmental stimuli. Examples of inducible genes would include those coding for activator and repressor proteins.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.1

17) What are sigma factors? Explain the roles of the two different sigma factors known as RpoH and RpoS.

Answer: A sigma factor (σ factor) is a protein needed only for initiation of RNA synthesis. The specific sigma factor used to initiate transcription of a given gene will vary, depending on the gene and on the environmental signals needed to initiate transcription of that gene. RpoH is the heat shock sigma factor and is turned on when bacteria are exposed to heat. RpoS is highly expressed during the transition from the exponential to stationary phase of the bacterial growth curve.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 6.10

18) What would be the result of a mutation that results in the inactivation of the Tar MCP in *Escherichia coli*?

Answer: The mechanism of chemotaxis depends upon a signal cascade of multiple proteins. These sensory proteins allow the cell to monitor the concentration of various substances over time and are called methyl-accepting chemotaxis proteins (MCPs). The Tar MCP produced by *Escherichia coli* senses aspartate and maltose and the repellants cobalt and nickel. Binding of an attractant or repellant triggers interactions with the cytoplasmic proteins that affect flagellar rotation. A mutation in the gene for Tar MCP would result in inability to move toward aspartate and maltose or away from cobalt and nickel.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.7

19) Explain how adaptation of the chemotaxis system is achieved once an organism has responded to a stimulus.

Answer: Once an organism has successfully responded to a stimulus, it must stop responding and reset the sensory system until a further signal is received. During adaptation of the chemotaxis system, a feedback loop that relies on the response regulator CheB resets the system. This is accomplished by methylation and demethylation of the MCPs by CheR and phosphorylated CheB respectively.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 6.7